INCH POUND

MIL-T-83815

26 Nov 90

MILITARY SPECIFICATION

TRUCK, TANK, LIQUID NITROGEN, AIRCRAFT SERVICING, 900 GALLON

This specification is approved for use within WR-ALC (CODE 84) *Department of the Air Force, and is available for use by all* Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a 900 gallon capacity truck used for transporting liquid nitrogen to a C-5 aircraft and servicing the Fire Suppression Systems with liquid nitrogen. The vehicle can deliver liquid and gaseous nitrogen to the aircraft using no auxiliary power or equipment.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

W-B-131	Battery, Storage (Vehicular, Ignition, Lighting and Starting)
VV-F-800	Fuel Oil, Diesel
KKK-T-2107	Trucks and Truck Tractors: Commercial, Diesel or Gasoline Engine Driven, 14,000 to 21,000 Pounds GVW, 4 X 2

MILITARY

MIL-P-514	Plates, Identification, Instructions and Marking, Blank
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4 and JP-5
MIL-W-16878	Wire, Electrical, Insulated High Temperature

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: WR-ALC/MMDETB,Robins AFB,GA. 31098-5609, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 2320

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MIL-P-23377	Primer Coating, Epoxy-Polyamide, Chemical and Solvent Resistant
MIL-C-46168	Coating, Aliphatic Polyurethane, Chemical Agent Resistant
MIL-C-53039	Coating, Aliphatic Polyurethane, Single Component, Chemical Agent Resistant
MIL-T-83133	Turbine Fuel, Aviation, Kerosene Type, Grade JP-8
MIL-R-83232/2C	Reel, Static Discharge, Grounding RMU 25/E 50 Foot with Connector
MIL-C-83286	Coating, Urethane Aliphatic Isocyanate, for Aerospace Applications

STANDARDS

FEDERAL

FED-STD-297	Rustproofing of Commercial (Nontactical) Vehicles
FED-STD-595	Colors

MILITARY

MIL-STD-12	Abbreviations for use on Drawings, and on Specifications, Standards and Technical Documents
MIL-STD-130	Identification Marking of US Military Property
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-721	Definition of Terms for Reliability and Maintainability
MIL-STD-781	Reliability Testing for Engineering Development, Qualification and Production
MIL-STD-808	Finishes, Materials and Processes for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Reports, Preparation of
MIL-STD-889	Dissimilar Metals
MIL-STD-965	Parts Control Program
MIL-STD-970	Standards and Specifications, Order of Precedence for the Selection of
MIL-STD-1223	Nontactical Wheeled Vehicle Treatment, Painting, Identification Marking and Data Plate Standards
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment and Facilities
MIL-STD-1556	Government/Industry Data Exchange Program (GIDEP), Contractor Participation Requirements
MIL-STD-1595	Qualification of Aircraft, Missile and Aerospace Fusion Welders
MIL-STD-1791	Internal Aerial Delivery in Fixed Wing Aircraft, Designing for

HANDBOOKS

MIL-HNBK-781	Reliability Test Methods, Plans, and Environments for Engineering Development, Qualification, and Production.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Ave., Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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LAWS AND REGULATIONS

49CFR393	Federal Motor Carrier Safety Regulations (FMCSR)
49CFR571	Federal Motor Vehicle Safety Standards (FMVSS)
49CFR172	Hazardous Materials Tables and Hazardous Materials Communications Regulations

(The Code of Federal Regulations (CFR) and the Federal Register (FR) are for sale on a subscription basis by the Superintendent of Documents, US Printing Office, Washington DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

EUROPEAN TYRE AND RIM TECHNICAL ORGANIZATION

STANDARDS

(Application for copies should be addressed to European Tyre and Rim Technical Organization, 32 Avenue, Brugmann-B, 1060 Brussells Belgium.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

J293	Vehicle Grade Parking Performance Requirements
J336a	Sound Level for Truck Cab Interior
J360	Motor Vehicle Grade Parking Performance Test Code
J366b	Exterior Sound Level for Heavy Trucks and Buses
J534	Lubrication Fittings
J695	Turning Ability and Off Tracking - Motor Vehicles
J786a	Brake System Road Test Code - Truck, Bus and Combination of Vehicles
J992b	Brake System Performance Requirements - Truck, Bus and Combination of Vehicles
J1077	Measurement of Exterior Sound Level of Trucks with Auxiliary Equipment
J1349	Engine Power Test Code, Spark Ignition and Diesel

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

TIRE AND RIM ASSOCIATION

YEAR BOOK

(Application for copies should be addressed to Tire and Rim Association, Incorporated, 3200 West Market Street, Akron OH 44313.)

COMPRESSED GAS ASSOCIATION (CGA)

Pamphlet S-1.1	Safety Relief Device Standards

(Application for copies should be addressed to Compressed Gas Association, Incorporated, 11 West 42nd Street, New York, NY.)

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DRAWINGS

MILITARY DRAWINGS

MS 17344R24N27S	Connector
8838302	Fastener Quality Assurance Requirements for WR-ALC/MMV Vehicle and Support Equipment
8991171	Coupling, Liquid Nitrogen

PARKER-HANNIFIN CORPORATION

7720016	Fill Coupling, Probe
7720021	Vent Coupling, Probe

CODES

ASME BOILER AND PRESSURE VESSEL CODE	Section VIII, Unfired Pressure Vessels

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

US DEPARTMENT OF AGRICULTURE (USDA)

5100-1A	Spark Arrestors for Internal Combustion Engines

(Application for copies should be addressed to US Department of Agriculture, Forest Service, Equipment Development Center, Dan Dimas, CA 91773.)

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Design.

3.1.1 Selection of specifications and standards. Specifications and standards, necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-970.

3.1.2 Materials. Materials shall be as specified herein. Materials not specifically covered by this specification or referenced document shall be suitable in every respect for this type document. Except as otherwise specified, wood shall not be used.

3.1.2.1 Protective treatment. Materials that are subject to deterioration shall be protected in such a manner that shall in no way prevent compliance with the performance requirements of this specification. Protective coatings that will chip, crack or scale with age or extremes of climatic conditions shall not be used.

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3.1.2.2 Dissimilar metals. Unless protected against electrolytic corrosion, dissimilar metals shall not be used in intimate contact with each other. Dissimilar metals are defined in MIL-STD-889.

3.1.2.3 Parts control. The contractor shall participate in Government Industry Data Exchange Program (GIDEP) in accordance with MIL-STD-1556B. In the design of the vehicle, the contractor shall ensure the components selected have not been identified as a problem or unresolved deficiency by GIDEP. The contractor shall submit an Alert/Safe-Alert, on a DD Form 1939, whenever an actual or potential problem is evident with respect to a part, component, material, test equipment, process or safety condition. The contractor shall certify maximum use of off-the-shelf items in accordance with MIL-STD-965A.

3.1.2.4 Recycled materials. All materials, equipment and articles incorporated in the work covered by this specification shall be new and fabricated using materials produced from recovered materials to the maximum extent possible without jeopardizing the intended use. The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification.

3.1.3 Design and construction. The vehicle shall be constructed so that parts do not work loose in service and shall withstand the loads, shocks, vibrations and other conditions incident to operation, shipping and storage with minimum loss of time for maintenance, repair and servicing. The design of the vehicle shall ensure safe, efficient and economical operation. All assemblies, controls and installed equipment shall be located so that there is no adverse interference with each other, or with the operation and shall be readily accessible for maintenance, operation, removal and replacement. All threaded steel fasteners of at least Grade 5 as well as any special threaded steel fasteners of a unique contractor design shall be verified as conforming to Air Force drawing 8838302.

3.1.3.1 Reliability. The vehicle, including all components, shall have a Mean-Time-Between-Failures (MTBF) of not less than 100 hours, demonstrated in accordance with Test Plan XXID of MIL-HDBK-781. A failure is defined as the event or inoperable state in which any item, or part of an item, does not, or would not, perform as previously specified. Definitions of reliability terms not included herein shall be as found in MIL-STD-721. The vehicle shall be designed for a life expectancy of at least 20 years.

3.1.3.2 Maintainability. The vehicle shall be designed and constructed to provide the following:

a. A minimum number of parts consistent with the performance required herein.

b. A minimum amount of time for assembling, disassembling, locating trouble sources and maintaining the vehicle. Where practical, components shall be located for rapid inspection and recognition of potential failure.

c. Ease of adjusting, servicing and replacing parts and components.

d. Use of readily available standard tools and equipment for maintenance.

e. Maintenance with a minimum number of tools.

f. Mean corrective maintenance downtime of no greater than one hour.

g. Maximum corrective maintenance downtime of no greater than four hours.

h. Maximum total downtime to perform all daily scheduled maintenance and servicing tasks of no greater than 35 minutes.

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3.1.3.2.1 Maintenance tasks. The vehicle shall be designed so that one mechanic using common hand tools can perform the following maintenance operations within the time specified:

a. Remove, replace and adjust all engine driven belts - .75 hour.

b. Remove and replace alternator - .5 hour.

c. Remove and replace all filters, screens and strainers in all hydraulic systems - one hour.

d. Remove and replace engine coolant system hoses - .5 hour.

e. Drain engine lubricating oil, remove and replace oil filter element(s) and refill crankcase-one hour.

f. Remove and replace fuel filter element (s) - .5 hour.

g. Disconnect battery cables, remove and replace batteries and reconnect battery cables - .5 hour.

h. Drain torque converter oil and transmission oil, remove and replace all filter elements and strainers and refill converter and transmission - one hour.

i. Remove and replace starter - one hour.

j. Remove or install the engine/transmission or engine/power divider package, two mechanics - four hours.

3.1.3.3 Foolproofness. Where improper installation of an item could cause a malfunction, an asymmetric mounting system shall be provided, where practical, to ensure proper mounting of the item.

3.1.3.3.1 Component removal. To the maximum practical extent, it shall be possible to remove components and major assemblies without the necessity of draining liquids or coolant, or dismantling other assemblies. Major assemblies shall be equipped with lifting eyes or other suitable provisions for easy attachment of lifting devices.

3.1.3.3.2 Cover plates. Cover plates, which must be removed for component adjustment or for component or part removal, shall be equipped with substantial quick disconnect fastenings.

3.1.3.4 Fastening devices. All screws, bolts, nuts, pins and other fastening devices shall be properly designed, manufactured and installed with adequate means of preventing loss of torque or adjustment. Cotter pins, lock washers or nylon patches shall not be used for this purpose, except for attachment of trim items or as provided in commercial (see 6.4.2) components. Tapped aluminum threads shall have a minimum thread engagement of at least two times the nominal fastener diameter.

3.1.3.5 Servicing provisions. Drains, grease fittings, lines and check points for all components shall be so located that they are readily accessible and do not require special tools for proper servicing.

3.1.3.6 Lubrication. All parts normally requiring grease for lubrication shall be provided with lubrication fittings in accordance with SAE J534. Where the use of high lubrication pressures may damage grease seals, a pressure relief device shall be provided. Extended fittings shall be provided to lubricate parts or assemblies which are not readily accessible for direct lubrication or which are likely to be overlooked because of inaccessibility.

3.1.3.7 Component protection. All space in which work is performed during operation, service and maintenance of the vehicle shall be free of hazardous protrusions, sharp edges or other features which

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may cause injury to personnel. All rotating and reciprocating parts and all parts subject to high operational temperatures or subject to being electrically energized, that are of such nature or so located as to be hazardous to personnel, shall be guarded or insulated to the extent necessary to eliminate the hazard.

3.1.3.8 Foreign object damage. All loose metal parts, such as pins and valve caps, shall be securely attached so they cannot become separated from the vehicle.

3.1.3.9 Bonding. Metal parts shall be specifically bonded together to minimize the possibility of sparks.

3.1.4 Environmental conditions. The vehicle shall be capable of satisfactory storage and operation as described herein under the following environmental conditions:

a. Full-time exposure to temperatures ranging from -40°F to +125°F for operation; -60°F to +160°F for storage.

b. Full-time exposure to relative humidities up to 100 percent.

c. Part-time (five percent of the time) exposure to rain; droplets of size 2.0 mm to 4.5 mm, falling at a rate of 13 cm per hour.

d. Part-time (five percent of the time) exposure to salt fog; five percent solution (five parts by weight of salt in 95 parts by weight of water) with fallout of 3.0ml/80cm^2/hour.

e. Part-time (five percent of the time) exposure to blowing dust; velocity 1,750 ft/min. concentration 0.5g/ft^3.

f. Part-time (five percent of the time) exposure to blowing sand; velocity 7,500 ft/min, concentration 0.0773g/ft^3.

3.1.5 Weight distribution. The weight distribution of the vehicle in the unloaded condition shall not be unbalanced or improperly distributed. The difference in load between the left and right wheels of any axle shall not exceed 2.5 percent of the total load on the axle.

3.1.6 Vehicle lean. The vehicle shall not lean to either side more than one percent of the maximum width, in either the fully loaded or the unloaded condition. Vehicle lean measurements shall be made at the outermost portion of the vehicle (excluding mirrors, mirror support brackets, lights or asymmetrical shapes) directly above each axle; the difference in body height at each axle shall not exceed one percent of the maximum width at that location.

3.1.7 Electromagnetic interference. The vehicle shall comply with the requirements of MIL-STD-461, Class C1, Group II for electromagnetic interference, except that the limits for UM03 (Figure 8-1) shall be relaxed by 20dB.

3.1.8 Sound levels.

3.1.8.1 Exterior sound level. The maximum sound level produced by the vehicle shall not exceed 83dBA when measured in accordance with SAE J366b. An additional 2dBA allowance shall be permitted for production units to provide for variation in test site, temperature gradients, test equipment and inherent differences in nominally identical vehicles.

3.1.8.2 Cab interior sound level. The maximum sound level in the cab interior shall not exceed the SAE recommended design criteria for new vehicles when measured in accordance with SAE J336a.

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3.1.8.3 Equipment operation sound level. The maximum sound level produced by the vehicle shall not exceed 85dBA when measured in accordance with SAE J1077. In addition, the vehicle shall be positioned so that the microphone is located 24 inches from the auxiliary equipment operator's control panel when tested by the stationary test procedure.

3.1.9 Air transportability. The vehicle shall be air transportable without the use of shoring in accordance with MIL-STD-1791 in Types C-130, C-141, and C-5 Aircraft and shall be air transportable in Type C-17 Aircraft. Preliminary design guidelines for Type C-17 are 142 inch maximum height, 204 inch maximum width, 812 inch maximum length, 9° ramp angle and floor load capacity equal to that allowed for the Type C-5 Aircraft.

3.1.9.1 Equipment removal. Removal of equipment shall be permitted, if necessary, to provide the required height and width for air transport. Removal and reinstallation of equipment shall be limited to removal and attachment of standard fasteners and fittings with common hand tools. The weight of individual parts or assemblies, which must be handled or positioned, shall not exceed the limits specified in MIL-STD-1472 for a one person lift. All equipment removed shall be stored on the vehicle; all necessary caps and plugs shall be provided to permit driving and storage of the vehicle while in the air transport configuration. Removal or reinstallation of the equipment shall take not longer than 30 minutes for two mechanics using common hand tools.

3.1.9.2 Tiedowns. The vehicle shall be equipped with sufficient tiedown points so located that it can be restrained in all directions during air transport. The attachment points shall be designed for use with attachment in the aircraft as described in MIL-STD-1791. Sufficient clearance shall be maintained around all attachment points to allow for restraining chains without interference from any other structure. The tiedown points shall have a minimum clear opening of three inches to allow attachment of a chain with a grabhook.

3.2 Performance.

3.2.1 Mobility. The fully loaded and equipped vehicle in the travel mode shall:

a. Maintain a speed of 60 miles per hour (MPH) on a level paved road

b. Accelerate from a stop to a speed of 45 MPH within 35 seconds on a level paved road.

c. Ascend and descend a 20 percent grade without spillage of any fluid.

d. Negotiate a 20 percent side slope in each direction.

e. Negotiate, in each direction, a ramp with a 30.6 percent slope connecting two horizontal surfaces without spillage of any fluid when configured for air transport. The ramp length shall be 20.0 feet. The vehicle shall be capable of being driven forward or backward from one horizontal surface across the ramp crest, ascend or descend the ramp, cross the ramp crest and proceed on the other horizontal surface without any portion of the vehicle, other than its tires, contacting any of the three surfaces.

f. Traverse graded gravel roads at a speed of 30 MPH.

g. Traverse Belgian Block at a speed of 20 MPH.

h. Traverse cross country terrain at a speed of 20 MPH.

i. Comply with all service brake performance requirements of SAE J992b.

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j. Comply with all parking brake performance requirements of SAE J293, except that the required grade shall be 20.4 percent.

k. Negotiate eight inch high blocks with diagonally opposite wheels of the vehicle.

l. Have a wall to wall turning diameter of 75 feet maximum in either direction.

3.3 Chassis. The chassis shall be the manufacturer's standard, except to comply with the specific requirements herein. The chassis shall conform to the requirements of KKK-T-2107, Type I.

3.3.1 Powertrain.

3.3.1.1 Engine. The vehicle shall be equipped with a commercial (see 6.4.2) liquid cooled diesel engine of not less than six cylinders. The total power output required by all items driven by the engine shall not exceed its net power output rating at 2,000 feet elevation, determined in accordance with SAE J1349. The engine shall demonstrate the performance characteristics specified herein when using diesel fuel in accordance with VV-F-800 and shall operate satisfactorily on Grades JP-4 and JP-5 turbine fuel in accordance with MIL-T-5624 and Grade JP-8 in accordance with MIL-T-83133. Any engine that is adversely affected by any of the above fuels is not acceptable.

3.3.1.1.1 Engine cooling system. The engine shall be equipped with a cooling system capable of satisfactory performance under the environmental conditions specified herein; coolant temperature shall not exceed the engine manufacturer's recommended maximum. Expansion tanks of sufficient capacity for expansion of all coolant in the system shall be provided.

3.3.1.1.2 Engine starting system. The engine shall be started within 15 seconds in any ambient temperature within the required operating range of the vehicle. Installed glow plugs, fluid starting aids and heat from the winterization system (see 3.6) may be used prior to and during the start period to facilitate engine starting under the following conditions:

TEMPERATURE RANGE (DEGREES F)	STARTING AIDS PERMITTED
40 through 125	None
1 through 39	Glow plugs and fluid starting aids
-40 through 0	Glow plugs, fluid starting aids and heat from the winterization system

The starting system shall be equipped with an interlock device to prevent starter engagement when the engine is running. A safety device shall be installed to prevent starting of the engine in any transmission selector setting other than neutral.

3.3.1.1.3 Exhaust system. The exhaust system shall be designed to expel exhaust gases in a manner that is not hazardous to operators or mechanics; the outlet shall not be directed toward the ground. The exhaust system, including all fasteners, shall be constructed of corrosion resistant materials. The system shall be protected from damage which could result from traversing rough terrain. A spark arrestor muffler conforming to USDA Standard 5100-1, shall be provided. This muffler shall be equipped with a clean out plug of at least 3/4 inch.

3.3.1.2 Transmission. A commercial (see 6.3.2) automatic transmission shall be provided, complete with a hydraulic torque converter. The normal drive range selector position shall provide not less than four gear ratios without selector movement. The transmission shall have a torque capacity of at least 110 percent of the maximum net engine torque.

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3.3.1.3 Axles. The load rating of each axle shall be not less than the imposed load on the axle, measured at the ground, with the vehicle at gross weight.

3.3.2 Steering system. The vehicle shall be equipped with a power assisted steering system, with manual steering available in the event of power failure. The steering system shall not require more than 4 turns lock to lock.

3.3.2.1 Steering wheel. The steering wheel shall have a diameter of 18 inches minimum. Not less than four inches clearance shall be provided around the steering wheel rim and the nearest projection in the plane of the steering wheel. Not less than , 2.5 inches clearance shall be provided between the steering wheel rim and the nearest projection below the steering wheel.

3.3.3 Brakes.

3.3.3.1 Service brakes. The vehicle shall be equipped with full air brakes on all wheels. Brakes shall be of either drum or disc type. A commercial (see6.4.2) off-the-shelf air dryer and necessary associated parts shall be added to the pneumatic system to eliminate excessive moisture and oil, thus increasing the life of the air brake components. It shall use desiccant for the final absorption process and shall be installed in the discharge line between the compressor and the first reservoir. A heater and thermostat assembly shall be provided with the air dryer to preclude freeze-up when used in severe winter conditions. A drain shall be provided at the lowest point on each air reservoir and shall be readily accessible for service.

3.3.3.2 Parking brakes. The vehicle shall be equipped with parking brakes of the spring applied type on all nonsteering wheels. A brake interlock override valve, to be used in emergency situations, shall be mounted in a protected location inside the cab, sealed in the "nonoverride" position.

3.3.4 Frame. The chassis frame shall be the manufacturer's standard for the type and class vehicle furnished. Chassis frame rails shall not project beyond the rear end of the body.

3.3.5 Suspension system. The vehicle shall be equipped with a suspension system with components having a rated capacity at least equal to the load imposed on each member measured at the ground with the vehicle loaded to specified GVW. When suspension capacity is rated at the spring pads, unsprung weight shall be deducted. The vehicle shall be equipped with hydraulic double acting shock absorbers at the front wheels.

3.3.6 Wheels and tires. The vehicle shall be equipped with wheels and tires which meet all Tire and Rim Association or European Tyre and Rim Technical Organization requirements for this application. Tubeless steel belted radial ply tires with highway type tread shall be provided.

3.3.7 Fuel system.

3.3.7.1 Fuel tank. A fuel tank shall be provided with a capacity of not less than 30 gallons. The tank shall have a fill opening of three inches minimum, readily accessible to personnel standing on the ground. It shall be located and mounted so as to provide maximum protection from damage.

3.3.7.2 Fuel filter. At least one filter of the fuel water separator type shall be located in the fuel line between the fuel tank and the engine. It shall be installed so that the element can be readily serviced without removing surrounding equipment. Unless the filter is mounted higher than the fuel tank, a shutoff valve shall be included in the inlet line to the filter.

3.4 Cab.

3.4.1 Mirrors. An adjustable rearview mirror of not less than 90 square inches shall be mounted on each side of the vehicle on a retractable bracket. The mirrors shall be positioned so that the driver can

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clearly observe the roadway and vehicles approaching from the rear on either side. A spotter mirror, at least five inches in diameter, shall be mounted below each rearview mirror. If a cab forward configuration is provided, a spotter mirror shall be mounted on each side of the windshield, installed so that the front of the vehicle is visible to the driver.

3.4.2 Instruments and controls. Instruments and controls shall be arranged so that those which are most vital to operation have the most predominant location. They shall be placed in blocks distinctly marked to group those instruments and controls pertinent to the operation of any individual component. For example, all instruments and controls relative to the operation of the engine shall be in one block. Controls having similar functions shall have similar actions. Indicator lights shall be green to indicate proper operation and red when warning is intended. All instruments and controls shall be illuminated with nonglare red lighting to be readily visible for night operation. At least the following instruments and controls shall be provided:

a. Speedometer and odometer.

b. Tachometer.

c. Engine oil pressure gauge.

d. Engine coolant temperature gauge.

e. Transmission oil temperature gauge.

f. Voltmeter or ammeter.

g. Fuel level gauge.

h. Engine hourmeter.

3.5 Electrical system. The vehicle shall be equipped with a 12 volt electrical system; a 24 volt starting system may be provided. Wiring shall be weatherproof and color coded or numbered for easy identification. All circuits shall be protected by circuit breakers located so as to be protected from the environmental conditions described herein and to be readily accessible to a mechanic.

3.5.1 Alternator. The engine shall be equipped with an alternator sufficient for operating all vehicle electrical components simultaneously. Minimum capacity of the alternator shall be 130 amperes.

3.5.2 Batteries. The vehicle shall be equipped with at least two 12 volt batteries, rated at a total of 1,800 CCA minimum. The batteries shall comply with the maintenance free characteristics of W-B-131. Batteries shall be so arranged and connected to provide sufficient capacity for engine starting in all conditions specified herein. The batteries shall be mounted in an acid resistant tray; a sliding carriage shall be provided if the batteries must be moved sideways to permit access for servicing or replacement. A master switch shall be provided to disconnect the batteries from the electrical system.

3.5.3 Lighting.

3.5.3.1 Vehicle lighting. The vehicle shall be equipped with headlights, taillights, parking lights, clearance lights, brake lights, turn signal lights and reflectors in accordance with Federal Motor Vehicle Safety Standard 49CFR571. The turn signal operating units shall have visible and audible flash indicators; turn signals shall be self-cancelling. Lights and reflectors mounted on the vertical surface of rub rails or on vehicle bumpers.

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3.5.3.2 Interior lighting. The cab and all compartments shall be provided with lighting adequate to enable personnel to readily obtain equipment and perform all functions necessary during night operation. Sealed contact switches shall be provided to automatically illuminate the lights in the cab or a compartment when its door is opened.

3.5.3.3 Night servicing light. A night servicing light shall be located on the left side of the truck. The light shall be fully adjustable with a switch located on or near the handle. It shall be located to illuminate the servicing area to allow safe efficient operation.

3.6 Winterization system. The vehicle shall be equipped with a winterization system to permit storage and operation in ambient temperatures as low as -40°F. Heaters shall operate on 110 volt alternating current (AC) and shall be wired through a junction block to a single three pronged (male) weatherproof slave receptacle for receiving external power and grounding the vehicle. A three wire connecting cable, 25 feet long and of adequate line capacity to supply power for all heater units simultaneously, shall be furnished. The connecting cable shall include a matching female connector at the vehicle end and a standard weatherproof three pronged (two power, plus one ground) male connector at the other end. Electrical insulation of connecting cable shall conform to FMCSR 393.77(c)(7). Electrical insulation of connecting cable shall withstand normal operating stresses in low ambient air temperature without cracking or loss of dielectric capacity. All heater lead wires shall be installed without interfering with vehicle component operation and without loose excess wire. Provisions for storage of the cable shall be provided in the vehicle cab. Heaters shall be furnished as follows:

a. Coolant heater, 1,500 watt minimum rating, shall be installed in the engine block or in the lower coolant inlet hose. Engine thermostat, with an operating range of 140°F to 160°F shall be in stalled.

b. Immersion type engine oil heater, 300 watt minimum rating, shall be installed in oil pan through any convenient opening. The oil heater shall be controlled by the coolant heater thermostat.

c. Battery heater shall be provided, having a capacity adequate to maintain battery electrolyte at a temperature of not less than 10°F during vehicle exposure in ambient air temperatures as low as -40°F and shall embody a thermostat to limit the temperature of the electrolyte to not more than 80°F.

d. A fuel warmer or preheater shall be provided to prevent clogging of fuel filters due to wax crystallization in the fuel. The fuel warmer shall use engine coolant to transfer sufficient heat to the diesel fuel to heat it from an inlet temperature of -40°F to an outlet temperature of +9°F, with a flow rate not less than the maximum fuel demand of the engine fuel system.

3.7 Functional design. The liquid nitrogen truck shall be a self-propelled flightline vehicle capable of receiving liquid nitrogen from base storage facilities, transporting and storing liquid nitrogen, service the C-5 aircraft liquid nitrogen fire suppression system, hot gas purge aircraft fire suppression system, fill and drain aircraft liquid nitrogen storage dewars and clear servicing lines into the vehicle's storage dewar or the atmosphere. The vehicle shall operate using existing aircraft servicing controls and power supplied by the vehicle. It shall be capable of being operated by one person. The vehicle shall be assembled of the manufacturer's standard commercial components to the maximum extent possible.

3.7.1 Major components.

a. Liquid nitrogen storage dewar (see 3.7.1.1).

b. Ambient air and exhaust gas heat exchangers (see 3.7.1.2 and 3.7.1.2.1).

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c. Fill and servicing lines (see 3.7.1.3 and 3.7.1.3.1)

d. Control panel and cabinet (see 3.7.1.4).

3.7.1.1 Storage dewar. The liquid storage tanks shall be a double wall vacuum insulated container. The tank shall consist of an inner and outer shell, inner shell suspension system, insulation, piping and connections, safety devices, controls and instruments necessary for meeting the requirements of this specification and its intended use. The insulation shall be aluminized polyester (mylar type). The inner vessel shall be fabricated and welded in accordance with Section VIII of the ASME Boiler and Pressure Vessel Code. The inner shell shall be certified under the code for a 90 psi of working pressure. The annular vacuum space shall be capable of maintaining a constant vacuum pressure for seven days when the inner shell is charged with 90 psi of nitrogen. Dissimilar metals shall not be used in the construction of the tank, piping and controls. All materials which come in contact with nitrogen shall be of the type which are unaffected by nitrogen or coated to protect them from nitrogen.

3.7.1.2 Heat exchangers. The heat exchanger shall be of sufficient size to supply heated gaseous nitrogen at a rate of 5.0 lb/min, plus or minus 0.5 lb/min, against a back pressure of 52 psig from the aircraft system. It shall also be capable of supplying gaseous nitrogen to the aircraft fill system at a rate of 0.021 lb/sec, plus or minus 0.003 lb/sec, against a 72 psig back pressure. The heat source shall be ambient air.

3.7.1.2.1 Purge heat exchanger. The purge gas heat exchanger shall use exhaust gas to heat nitrogen at a flow rate of 5 lb/min to 180°F. The heat exchanger and all associated hardware shall be constructed of stainless steel. If shall be bolted or clamped to other exhaust components as to allow easy removal and reinstallation.

3.7.1.3 Fill and servicing hoses. The aircraft to truck servicing hoses shall be a 17 foot flexible hose attached to rigid piping with wrought union copper to copper (or stainless steel), inside diameter 1.375 inch X 1.25 inch for fill line and 2 inch X 2.5 inch for vent line, on the left side of the vehicle aft of the cab. The flexible hose and piping shall be compatible with the material carried in the storage tank. The servicing hoses shall have a 90° elbow with a 360° swivel at each end. One hose shall terminate with a fill probe, per Parker Hannifin Drawing 7720016-103. The other shall have a vent probe, per Parker Hannifin Drawing 7720021-103. Provisions shall be made for their storage on the vehicle when not in use.

3.7.1.3.1 Ground storage connection. The vehicle shall have a coupling in accordance with Ordinance Drawing 8991171 located on the left side of the vehicle aft of the cab. The coupling shall be used to fill the storage dewar from ground to storage tanks.

3.7.1.3.2 Aircraft electrical connection. A shielded eight conductor electrical cable 19 feet long conforming to MIL-W-16878 shall be provided, terminating in a connector in accordance with MS 17344R24N27S. The cable shall be enclosed in a scuffproof conduit attached to the vehicle on the left side of the vehicle aft of the cab. A storage rack shall be provided for the cable.

3.7.1.4 Control panel. A control panel shall be located in a cabinet on the left side of the truck aft of the cab. The cabinet and hardware shall be zinc plated. The panel shall contain all controls and instruments required for normal operation and maintenance of the mounted equipment. The controls and gauges shall be located so they are easily readable and operational simultaneously by one person standing on the ground in front of the panel. The control panel shall be illuminated for night operation. The lighting shall be designed to illuminate all gauges and controls. Controls shall be operable by personnel wearing arctic mittens.

3.7.1.4.1 Gauges. All gauges shall be of a size which is easy to read while standing in front of the control panel. All gauges shall be clearly marked as to their function. The scale on each gauge shall be marked in increments sufficient to monitor all operations accurately. All gauges shall have orifices or

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snubbing devices to control fluctuations of the indicator. All gauges shall be preceded by a valve which will allow removal of the gauge without depressurizing the system.

3.7.1.4.2 Compartment and racks. Compartments and racks shall be provided for storage of all equipment. Compartments shall protect their contents from water or cleaning solvents. Drain holes shall be provided in each section of the compartment. The compartments shall have access doors of sufficient size to allow operation, servicing and removal of equipment. Doors shall have devices to securely hold them in the open position while operating. The hinges, latches and compartments shall be sturdy to withstand the dynamic loads imposed by operating the vehicle. All latches and hinges shall be attached by bolts to allow easy replacement. Doors shall be designed to be operable under all weather conditions by an operator wearing arctic mittens without the need of any tools.

3.7.1.4.3 Wheel check storage rack. A 12 inch by 24 inch by 10 inch deep chock storage rack shall be provided. It shall be located in or next to the control panel.

3.7.1.4.4 Handbook and record compartment. A 9 inch by 12 inch by 2 inch enclosure shall be provided in addition to the vehicle glove compartment for storage of a record and handbook. It shall be weatherproof or in a weatherproof location.

3.7.1.4.5 Servicing hose storage. Storage for all servicing hoses shall be provided. It shall be designed to allow easy removal of hoses, but prevent loss or damage to hoses, nozzles or probes during storage and transportation.

3.7.1.5 Valves and piping. All valves and piping shall be provided in spool type assemblies, assemblies which are threaded together and are removable with common wrenches. The valves shall provide positive shutoff when closed and minimum resistance to flow when fully opened. All valves shall have renewable type disc or seals. The valve stems shall be made of low heat conductive material. The handles shall be color coded as follows:

VALVE	COLOR
Fill Line Bypass Shutoff	Dark Blue
Saturation Line Bypass Shutoff	Yellow
Storage Pressure Relief Shutoff	Red
Fill Line Vent Shutoff	Red
Dewar Vent Shutoff	Red
Heat Exchanger Shutoff	White
Ground Storage Shutoff	Green
Purge Line Shutoff	Gray

All materials used in the construction of the valves and piping shall be unaffected by liquid nitrogen.

3.7.1.5.1 Line strainer. The fill line for filling the vehicle storage tank shall be equipped with a Y-type strainer designed for a minimum flow rate of 25 GPM with a minimal pressure drop. The strainer shall be a minimum of 100 mesh and be easily removable for cleaning.

3.7.1.6 Cleaning solvent drains. Provisions shall be made to ensure cleaning solvents and flushing liquids can be thoroughly removed from the liquid nitrogen system without disassembly. The use of threaded drain plugs, where it is impractical to provide drain valves, is acceptable. All cabinets and storage shall have provisions for drainage.

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3.7.1.7 Full level trycock. The storage dewar shall be equipped with a full level trycock line.

3.7.2 Performance.

3.7.2.1 Storage tank. The vehicle shall receive liquid nitrogen from ground storage tanks. The nitrogen shall be filtered before entering the tank and enter the top of the inner vessel through a spray header. A shutoff valve shall be provided for venting the tank during filling. A gauge shall be provided to indicate the quantity of nitrogen in the tank. All necessary relief valves and burst disc shall be provided. The tank shall have a gross volume of 900 gallons and usable volume of 810 gallons. The fill rate from the ground storage tanks to the vehicle's nitrogen tank shall be 15 GPM at 20 psig through a 15 foot transfer hose. Nitrogen losses from the vehicle's system shall not exceed 1.5 percent of its net capacity over a 24 hour period when the vehicle is pressurized to 20 psig.

3.7.2.2 Aircraft servicing. The vehicle shall be capable of performing all servicing functions without external power. The vehicle shall supply 24 volts, plus or minus 6 volts, direct current at 2.0 amperes to the aircraft.

3.7.2.2.1 Purge and fill. The vehicle shall supply heated (180°F) gaseous nitrogen to the aircraft at a rate of 5.0 lb/min, plus or minus 0.5 lb/min, at a 52 psig back pressure. The truck shall saturate the aircraft system with gaseous nitrogen at a transfer rate of 0.021 lb./sec, plus or minus 0.003 lb/sec, against a 72 psig back pressure. It shall be capable of supplying liquid nitrogen at a rate of not less than 20 GPM with 75 psig of back pressure.

3.7.2.2.2 Drain and clear. The vehicle shall be capable of draining the aircraft nitrogen system and clearing the aircraft and vehicle fill lines of nitrogen by venting them to the atmosphere.

3.7.2.2.3 Vent and pressure line. The flow resistance of the vent and pressure line shall be the minimum practical for this type of equipment. The vent line shall include a manual valve to restrict back pressure to utilize the temperature monitor which shall be installed upstream of the valve during purging. The discharge line transfer hose shall be jacketed with 0.5 inch flexible foam insulation, or equal, having a K factor of 0.15 BTUs per foot square, degree Fahrenheit, inch of thickness. All rigid lines that carry liquid nitrogen shall have a minimum of 0.75 inch of the insulation described above. All hoses and lines shall be resistant to scuffing and abrasions normally encountered during operation.

3.7.2.2.4 Maximum transfer pressure. There shall be no operation condition where the vehicle can supply over 90 psig to the aircraft. The vehicle shall be designed so that in the event of a component failure, the vehicle cannot possibly supply more than 135 psig to the aircraft.

3.7.2.3 Self purge. The vehicle shall be capable of self-purging when connected to base supply facilities utilizing the exhaust gas heat exchanger.

3.7.2.4 Static grounding. The vehicle shall have a static ground reel in accordance with MIL-R-83232/2 located on the right rear of the truck.

3.7.2.5 Safety interlock. The vehicle shall be equipped with a connector type holder which retains the fill and vent probes when they are not in use. The interlock shall insure that the vehicle parking brakes are applied when either the fill, vent, or both probes are removed from the holder.

3.7.2.5.1 Pressure safety devices. Pressure limiting devices shall be provided to prevent dangerous pressures from developing in any part of the system which would cause damage to any component or injury to personnel. The vehicle shall be designed to meet the human engineering requirements of MIL-STD-1472.

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3.7.2.6 Gauges and valves. All gauges and valves required to monitor temperature and pressure during operation of the liquid nitrogen system of the vehicle shall be provided. They shall be located on the control panel.

3.8 Finish and protective coatings.

3.8.1 Cleaning, painting, plating, anodic films and chemical treatments. Cleaning, painting, plating, anodic films and chemical treatments shall be in accordance with MIL-STD-808. Unless otherwise approved by the procuring activity, primer shall be in accordance with MIL-P-23377.

3.8.1.1 Finish.

3.8.1.1.1 Forest Green. Unless otherwise specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint, Forest Green, Color Number 24052 of FED-STD-595. The paint shall be in accordance with MIL-C-83286, or shall be DuPont Imron, or equal. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, light trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

3.8.1.1.2 Desert Sand. When specified (see 6.2), the outer surface of the vehicle and the interior surfaces of compartments (except in the cab) shall be painted with polyurethane paint in accordance with MIL-C-46168 or MIL-C-53039, Desert Sand, Color Number 30313 of FED-STD-595. All exterior trim items (such as wheels, wheel covers, running boards, metal radiator grilles, bumpers, light trim rings, door handles and drip rails) shall be painted with the same paint as the exterior of the vehicle.

3.8.1.1.3 Rustproofing. Unless otherwise specified (see 6.2), the vehicle shall be rustproofed in accordance with FED-STD-297. The first article units shall not be rustproofed until after approval of the first article test report.

3.8.2 Markings. Unless otherwise specified (see 6.2), marking for vehicles painted Forest Green shall be applied with black paint, Color Number 37038 of FED-STD-595, and marking for vehicles painted Desert Sand shall be applied with white paint, Color Number 37875 of FED-STD-595. Each end and side of the truck shall be marked with a "NONFLAMMABLE GAS" placard as required by 49CFR172.

3.8.2.1 Identification markings. Unless otherwise specified (see 6.2), the vehicle shall be marked for the appropriate service in accordance with MIL-STD-1223.

3.8.3 Identification plates and operating instructions. All identification plates, instruction plates, data plates, etc, shall conform to MIL-P-514 and be of a size and shape consistent with the information required thereon. Abbreviations shall be in accordance with MIL-STD-12.

3.8.3.1 Identification plate. An identification plate, permanently marked in accordance with MIL-STD-130, shall be securely attached to the vehicle in a readily accessible location. The identification plate shall contain the following information:

NOMENCLATURE

MANUFACTURER'S MODEL NUMBER

MANUFACTURER'S SERIAL NUMBER

MANUFACTURER'S NAME

MANUFACTURER'S ADDRESS

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MANUFACTURER'S SERVICE PHONE NUMBER

DATE OF MANUFACTURE

CONTRACT NUMBER

NATIONAL STOCK NUMBER

3.8.3.2 Control markings. All controls, valves, gauges and indicators used in the operation of the vehicle shall be identified by securely attached nameplates.

3.8.3.3 Operating instructions. Brief operating and precautionary instructions shall be permanently affixed near the appropriate controls. Operating instruction plates and nitrogen system schematics shall be provided in the control compartment. The instructions shall be clear, concise and adequate to enable operation of the vehicle without damage to the equipment or injury to personnel and refer to the components as identified by nameplates.

3.8.3.4 Lubrication plate. A lubrication plate shall be provided directing attention to all lubrication fittings and components which require lubrication. The plate shall identify the type and grade of lubricant required for all operational temperatures.

3.8.3.5 Transportation data plate. A transportation data plate shall be provided. The plate shall contain at least the following information:

a. Side and rear silhouette views of the vehicle.

b. Horizontal and vertical location of the center of gravity of the vehicle in transport condition.

c. Shipping weight.

d. Loading cubage.

e. Overall length, width and height.

f. Tiedown instructions for air transport in Types C-130, C-141, C-5, and C-17 Aircraft.

3.8.4 Part numbering of interchangeable parts. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. All parts shall be marked for identification in accordance with MIL-STD-130. (Note that MIL-STD-130 does not require marking of commonly known commercial (see 6.4.2) parts which present no identification problem or parts in assemblies which are not normally subject to disassembly or repair.)

3.9 Welders and welding.

3.9.1 Welders. All welders employed in the fabrication of the vehicle, except for commercial (see 6.3.2) components, shall be certified in accordance with MIL-STD-1595, the American Welding Society or the American Society of Mechanical Engineers. The certification shall be on file at the contractor's facility and available for review by the procuring activity.

3.9.2 Welding. Welding procedures shall be in accordance with a nationally recognized welding code. The surface parts to be welded shall be free from rust, scale, paint, grease and other foreign matter. Welds shall be of sufficient size and shape to develop the full strength of the welded parts. Welds shall transmit stress without cracking or permanent distortion when the parts connected by the welds are subjected to test, proof and service loadings.

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3.10 Connections.

3.10.1 Bolted connections. Bolt holes shall be accurately punched or drilled and shall be deburred. Unless torque is specified, threaded fasteners shall be tight and shall not work loose during testing or service usage.

3.10.2 Riveted connections. Rivet holes shall be accurately punched or drilled and shall be deburred. Rivets driven with pressure tools shall completely fill the holes. Rivet heads, when not countersunk or flattened, shall be of approved shape. Rivet heads shall be full, neatly made, concentric with the rivet holes and in full contact with the surface of the compartment.

3.11 Gear and lever assemblies. Gear and lever assemblies shall be properly aligned and meshed and shall be operable without interference, tight spots, loose spots or other irregularities. Where required for accurate adjustment, gear assemblies shall be free of backlash.

3.12 Cleaning. The vehicle shall be thoroughly cleaned. Loose, spattered or excess solder, welding slag, stray bolts, nuts and washers, rust, metal particles, pipe compound and other foreign matter shall be removed during and after final assembly.

3.13 First Article. A first article sample shall be subjected to first article inspection (see 6.5) in accordance with 4.4.

3.14 Workmanship. The vehicle, including all parts and accessories, shall be constructed and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs and sharp corners and edges; accuracy of dimensions, surface finish and radii of fillets; thoroughness of soldering, welding, brazing, painting, wiring and riveting, marking of parts and assemblies; alignment of parts and tightness of assembly fasteners, etc.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of tests. The inspection and testing of the vehicle shall be classified as follows:

a. First article testing (see 4.4).

b. Acceptance testing (see 4.5).

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4.3 Test conditions.

4.3.1 Apparatus. Apparatus used in conjunction with the testing specified herein shall be of laboratory precision type and shall be calibrated at proper intervals to ensure laboratory accuracy.

4.3.2 Data. During all testing specified herein, at least the following data, unless not applicable, shall be recorded at intervals not to exceed 30 minutes. Additional data and/or shorter intervals shall be provided as appropriate for any specific test.

a. Date.

b. Time started.

c. Time finished.

d. Ambient temperature.

e. Engine speed.

f. Engine oil pressure.

g. Engine coolant temperature.

h. Transmission oil temperature.

i. Vehicle odometer reading at the start and finish of each test.

j. Engine hourmeter reading at the start and finish of each test.

4.3.2.1 Operational data. Operational data shall be recorded at the intervals required by the various test procedures. The data recorded shall include the following, as applicable to the particular test being performed:

a. Liquid storage tank assembly:

(1) Inner shell vapor phase pressure (psig).

(2) Liquid level (pounds).

b. Liquid servicing system:

(1) Pressure on liquid leaving servicing hose (psig).

(2) Servicing flow rate (GPM).

4.3.3 Preparation for test. The vehicle shall be prepared for test by assemblies, adjusting and servicing it in accordance with the operating and maintenance instruction handbook. The servicing shall include vacuum pumping of the liquid storage tank annular insulation space and sealing of the vacuum line.

4.3.3.1 Vacuum pumping. A vacuum pump shall be connected to the liquid storage tank of the vacuum line disconnect fitting and the annular insulation space evacuated as necessary for compliance with the performance requirements of 3.7.1.1.

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4.3.3.2 Vacuum line seal. Upon completion of the vacuum pumping, the vacuum pump shall be disconnected and the vacuum line so sealed that further vacuum pumping cannot be accomplished without breaking the seal. The vacuum line shall remain sealed throughout all testing specified herein.

4.3.4 Test rejection criteria. Throughout all tests specified herein, the vehicle shall be closely observed for the following conditions, which shall be cause for rejection:

a. Failure to conform to design or performance requirements specified herein.

b. Spillage or leakage of any liquid, including fuel, coolant, lubrication or hydraulic fluid, under any operating condition, except as allowed herein.

c. Structural failure of any component, including permanent deformation, or evidence of impending failure.

d. Evidence of excessive wear.

e. Interference between the vehicle components or between the vehicle, the ground and all required obstacles, with the exception of normal contact by the tires.

f. Misalignment of components.

g. Evidence of undesirable roadability characteristics, including instability in handling during cornering, braking and while traversing rough terrain.

h. Conditions which present a safety hazard to personnel during operation, service or maintenance.

i. Overheating of the engine, transmission or any other vehicle component.

4.4 First article testing.

4.4.1 First article test samples. The first article test samples shall be identical to the production items procured under the terms of the contract. Approval of the first article samples shall not constitute a waiver by the Government of its rights under the provisions of the contract. Two first article samples shall be produced and tested as follows:

a. Contractor test sample (see 4.4.2.1).

b. User test sample (see 4.4.2.2).

4.4.2 First article tests.

4.4.2.1 Contractor test sample. The contractor test sample shall be subjected to the tests described in 4.6.1 through 4.6.9.

4.4.2.2 User test sample. The user test sample shall successfully complete the tests in 4.6.1, 4.6.2, 4.6.4.2 and 4.6.4.3 prior to being shipped to the user test facility for the test described in 4.6.9.

4.4.3 Notification of tests. The procuring activity shall be notified of the time and location of all first article tests so that representatives of the procuring activity may attend. The notification shall be in writing and shall include the nomenclature and the national stock number (NSN).

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4.4.4 Test report. The following shall be incorporated as a part of a test report:

a. All data collected, failures, maintenance performed and other events recorded shall be identified by accumulated operation time, miles, cycles or position of the test procedures, as appropriate. Test conditions at the time of failures or irregular operations identified shall be recorded.

b. Summary of the engineering reasoning and of any tests conducted to determine assignable causes for all failures and irregular operations identified.

c. Summary of the engineering reasoning behind any corrections made, to be made or proposed to be made and the predicted effectiveness of those corrections.

d. Test activity and/or contractor comments on item features or requirements that, if modified, should improve the item.

e. Test activity and/or contractor comments on field conditions or operating procedures to be avoided or cultivated to increase the reliability and useful life of the item.

f. List of all cleaning and finishing procedures to be used in accordance with MIL-STD-808.

g. List of methods to be used to protect against electrolytic corrosion of dissimilar metals in intimate contact in accordance with MIL-STD-889.

h. List of special tools and diagnostic equipment necessary to perform field level maintenance.

i. Plan for testing production units.

4.5 Acceptance testing. Acceptance testing shall be classified as follows:

a. Individual testing (see 4.5.1).

b. Random sample testing (see 4.5.2).

4.5.1 Individual testing. Each production vehicle shall be subject to the tests specified in 4.6.1, 4.6.2, 4.6.4.2 and 4.6.4.3.

4.5.2 Random sample testing. Any production vehicle may be selected by the procuring activity at any time to be subjected to the tests described in 4.6.1 through 4.6.6.6 and 4.6.7. Upon completion of the random sample testing, the contractor shall prepare a test report in accordance with MIL-STD-831. An authorized Government representative shall countersign all data sheets of the report.

4.5.2.1 Random sample test failure. When a vehicle fails to successfully complete any random sample test, the cause of the failure shall be determined, corrective action shall be taken and the test shall be repeated to verify adequacy to the corrective action. Tests not involving the defective component(s) may be continued pending the investigation of a test failure. Vehicles to be delivered pursuant to the contract shall be inspected for all defects identified during all random sample tests and may not be accepted prior to verification of adequacy for the corrective action for a random sample test failure.

4.6 Test methods.

4.6.1 Examination of product. The vehicle shall be inspected to determine compliance with the requirements specified herein. Particular attention shall be given to materials, workmanship, dimensions,

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surface finishes, protective coatings and their application, welding and riveting and markings. For the first article and random sample vehicles, this examination shall be accomplished by use of a checklist which lists each requirement not validated by a test and the results of the examination.

4.6.2 Preliminary operation.

4.6.2.1 Pressure test. Each component and circuit of the vehicle that operates under positive pressure in normal service shall be pressure tested to 1.5 times its normal working pressure. Automatic pressure relief devices need not be installed for this test. Any distortion, damage or leaks resulting from the pressure testing shall be cause for rejection of the components affected.

4.6.2.2 Bursting disc safety device sampling and test. Three bursting disc safety devices of each type (same size, design, construction and pressure) from each lot of 100 or less shall be selected at random and subjected to the testing required by CGA Pamphlet S-1.1. Should any sample bursting disc safety device fail to satisfactorily pass the specified testing, five additional sample bursting disc safety devices from the same lot shall be tested in the same manner.

4.6.2.2.1 Bursting disc safety device lot. A lot shall consist of bursting disc safety devices manufactured under essentially the same conditions and submitted for inspection and substantially the same time.

4.6.2.2.2 Bursting disc safety device rejection and retest. If more than one of the three original samples, or any of the five additional bursting disc samples, fail at a pressure outside the limits specified, the entire lot shall be rejected.

4.6.2.3 System functional checkout. The vehicle shall be tested to verify the functional capability of the valves, pressure gauges, solenoid valves, liquid level gauges and relief valves.

4.6.2.3.1 Safety interlocks. Demonstrate that the parking brakes cannot be released unless the fill and vent probes are stored in their holders or the brake override valve is activated.

4.6.2.3.2 Electrical test. Each unit shall be subjected to the following verifications: Each electrical subassembly and final assembly shall be tested for continuity using a volt ohmmeter. The vehicle to aircraft electrical power shall be verified to meet the requirements of 3.5.2.

4.6.2.3.3 Leakage test. Pressurization of the completed vehicle to high and low operating pressures shall be performed to visually verify that no leakage exists in the storage vessel or piping assemblies. Any evidence of leakage shall be cause for rejection of the unit.

a. Low pressure test - 90 psig, plus or minus 0 psig.

b. High pressure test - 135 psig, plus or minus 5 psig.

4.6.2.4 Road handling test. The vehicle shall be filled with liquid nitrogen and driven 25 miles at approximately 45 MPH. The vehicle shall be driven through at least five short radius turns to the left and an equal number to the right and at least five sudden stops.

4.6.3 Mobility tests.

4.6.3.1 Road tests. The truck shall be driven 2,000 miles consisting of five cycles of 400 miles each. The first cycle shall be run with the truck empty. The second and third cycles shall be run with the truck approximately half loaded. The fourth and fifth cycles shall be run with the truck loaded to rated capacity. During the highway portion of each cycle, at least 20 short radius right turns, 20 short radius left turns and

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10 sudden stops shall be accomplished. Each cycle shall be performed continuously without rest and shall be performed on these surfaces for the distances and speeds listed below:

a. Highway - 200 miles at an average speed of at least 45 MPH.

b. Gravel road - 100 miles at an average speed of at least 25 MPH.

c. Belgian Block pavement - 100 miles at an average speed of at least 10 MPH.

4.6.3.2 Maximum speed and acceleration. The time speed distance relationship of the fully loaded vehicle shall be determined while accelerating from a standing start to maximum speed on a dry level paved surface to demonstrate compliance with 3.2.1(a) and 3.2.1(b). Not less than three runs, each in opposite directions, shall be conducted. Maximum speed and acceleration shall be the average for all runs.

4.6.3.3 Grade operation. The fully loaded vehicle shall be driven up and down a 20 percent dry paved grade. The maximum speed that can be maintained while climbing the grade shall be measured and recorded in the test report. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended minimum pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the grade. Vehicle shall be observed to demonstrate compliance with 3.2.1(c).

4.6.3.4 Side slope operation. The fully loaded vehicle shall be driven in both directions on a side slope of 20 percent. Engine oil pressure shall be observed and shall not be less than the manufacturer's recommended pressure at the engine operating speed. Engine starting ability shall be demonstrated in both directions on the slope.

4.6.3.5 Ramp test. The fully loaded vehicle shall be driven forward from a hard flat horizontal surface and ascend a ramp with a minimum length of 1.2 times the vehicle wheelbase and a slope of 30.6 percent. It shall then proceed across the ramp crest onto an upper connecting hard horizontal surface until all tires are resting on the upper surface, down the ramp, and onto the lower horizontal surface. The test shall then be repeated in the opposite direction.

4.6.3.6 Suspension articulation. The vehicle shall be tested to demonstrate compliance with 3.2.1(k). The side of the blocks may be sloped.

4.6.3.7 Wall to wall turning diameter measurement. The vehicle wall to wall turning diameter for turns in each direction shall be measured in accordance with SAE J695 to demonstrate compliance with 3.2.1 (1).

4.6.3.8 Brake tests.

4.6.3.8.1 Service brake tests. The vehicle shall be tested in accordance with SAE J786a to demonstrate compliance with 3.2.1(i).

4.6.3.8.2 Parking brake tests. The vehicle shall be tested in accordance with SAE J360 to demonstrate compliance with 3.2.1(j).

4.6.4 Servicing test.

4.6.4.1 Discharge line test. The vehicle shall be operated in the fill mode. The flow requirements of 3.7.2.2.1 shall be demonstrated.

4.6.4.2 Heat exchanger test. The vehicle shall be operated in the purge and saturation modes. The flow requirements of 3.7.2.2.1 and the temperature requirements of 3.7.1.2 shall be demonstrated.

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4.6.4.3 Servicing capability test. The vehicle liquid and gaseous nitrogen transfer capability of 3.7.2.2.1 shall be demonstrated.

4.6.4.4 Vehicle fill. The vehicle's liquid storage tank shall be filled from a liquid nitrogen ground storage tank using the liquid nitrogen ground storage tank manufacturer's recommended tank pressure in accordance with the operating instructions specified for pressure transfer in technical orders (operation and maintenance instruction handbook).

4.6.4.5 Aircraft fill and vent. With the liquid fill and gas vent servicing hoses connected to an actual, mockup, or equivalent, aircraft fill and vent system, fill the aircraft dewar with not less than 750 pounds of liquid nitrogen. The liquid nitrogen fill rate shall be in accordance with 3.7.2.1.

4.6.4.5.1 Aircraft fill line clear. The aircraft servicing line shall be cleared. The final pressure in the servicing line, upon completion of the clearing, shall not exceed 20 psig.

4.6.4.5.2 Aircraft saturation. The liquid nitrogen transferred above shall be saturated between 40 psig to 50 psig at a rate of not less than 1.0 psig per minute.

4.6.4.5.3 Aircraft drain. The liquid nitrogen shall be drained from the receiving vessel.

4.6.4.6 Vacuum retention test. The liquid storage tank inner shell shall be pressurized to not less than 90 psi with nitrogen gas. The vacuum line shall be sealed to prevent further evacuation during the test and the liquid storage tank annular insulation space absolute pressure measured. The liquid storage tank shall remain in this condition for seven days minimum; then the annular insulation space absolute pressure shall be determined.

4.6.4.7 Heat leak evaporation loss rate test. The heat leak evaporation loss rate test shall consist of the following individual test operations conducted in the order listed. Throughout the complete test, the liquid storage tank inner shell vapor phase pressure shall remain at, or as near, atmospheric pressure as the measuring apparatus will permit.

4.6.4.7.1 Liquid storage condition heat leak evaporation loss rate test. Following the vacuum retention test specified under 4.6.4.6, the liquid storage tank inner shell shall remain filled to not less than 50 percent of its design capacity with liquid nitrogen. The liquid storage tank shall then be permitted to temperature stabilize until a constant annular insulation space absolute pressure is obtained.

4.6.4.7.2 Liquid storage tank capacity test. The liquid storage tank assembly storage and transport capacity shall be determined by filling a liquid into the liquid storage tank inner shell with the vehicle parked on a flat level surface until the filling liquid overflows through the full trycock line, recording the liquid storage tank assembly liquid level gauge reading in terms of liquid nitrogen, and then either measuring the volume of the filling liquid or determining its volume from weights before and after filling. The liquid storage tank inner shell shall then be filled completely to determine total internal volume, and the quantity of filling liquid measured as above. The actual liquid volume when the liquid storage tank is filled to overflow through the vapor vent line and when completely filled shall both be recorded in US gallons and the data shall be accurate to within plus or minus 0.5 percent of gross capacity. The liquid storage tank actual liquid contents when liquid starts to discharge through the vapor vent line and when the liquid storage tank is completely filled shall be at least 810 gallons.

4.6.4.7.3 Venting. The inner shell vapor space shall be vented to atmospheric pressure through the flow measuring apparatus that will be used for determining the evaporation loss rate. It shall remain under the specified ambient temperature and liquid storage tank inner shell vapor phase pressure conditions for 24 hours while heat leak evaporation rate losses for the liquid storage tank are determined by passing all vapors vented from the inner shell through a recording, or totaling, type case flowmeter. The total weight of vapors vented during the test and the average barometric pressure to which the vehicle has been

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subjected during the period shall be recorded at intervals of not to exceed four hours during the 24 hour test. The actual rate of loss for each reading, the average actual loss over the entire 24 hours test period and the average ambient temperature to which the vehicle was exposed during the 24 hour test period shall be calculated. The evaporation loss rate shall be reported as the pounds of liquid nitrogen vaporized per 24 hour day and shall not exceed the limit specified in 3.7.2.1.

4.6.5 Design verification.

4.6.5.1 Weight and weight distribution. The weight on each corner of the vehicle shall be measured with the unit standing on a flat level surface, both in the empty and the fully loaded conditions to demonstrate compliance with 3.1.5. The fully loaded vehicle shall include ballast to simulate a 250 pound crew member in each seating position.

4.6.5.2 Dimensional verification. The dimensions specified herein shall be verified with the vehicle standing on a flat level surface. Overall height and vehicle lean shall be measured in both the empty and the fully loaded configurations, as defined in 4.6.5.1.

4.6.5.3 Electromagnetic interference measurement. The vehicle and all electrical components shall be tested in accordance with MIL-STD-462 to demonstrate compliance with 3.1.7.

4.6.5.4 Sound level testing. The vehicle shall be tested in accordance with SAE J336a, J366b and J1077 to demonstrate compliance with 3.1.8.

4.6.6 Environmental tests.

4.6.6.1 Low temperature storage and operation. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph 4.6.6.5. Engine starting aids may be used (see 3.3.1.1.2). A fine mist of water shall be applied to the entire windshield to form a thin film of ice. The defroster shall clear the windshield within one hour and maintain clear vision throughout the remainder of the test. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the minimum required cold storage temperature of 3.1.4(a) in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the minimum required cold operational temperature of 3.1.4(a) in a steady state condition.

4.6.6.2 High temperature storage and operation. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 502.2, Procedures I and II, of MIL-STD-810 to demonstrate compliance with 3.1.4(a). Temperature sensor locations shall be approved by the procuring activity. Operational checkouts shall be performed in accordance with paragraph 4.6.6.5. "Standard ambient" and "controlled ambient" shall be defined as the ambient condition of the test site. For Procedure I, the vehicle shall be exposed to the maximum required hot storage temperature of 3.1.4(a) in a steady-state condition for 24 hours. For Procedure II, the vehicle shall be exposed to the maximum required hot operational temperature of 3.1.4(a) in a steady-state condition.

4.6.6.3 Rain. The complete vehicle, including all accessories, shall be tested in accordance with Test Method 506.2, Procedure III, of MIL-STD-810, to demonstrate compliance with 3.1.4(c). Operational checkouts shall be in accordance with 4.6.6.5. The interior of the cab and all compartments shall be examined for moisture; any evidence of leakage shall be cause for rejection.

4.6.6.4 Other environmental tests. The components and materials used in the construction of the vehicle shall be subjected to, or certified as having passed, or being capable of passing, the following test methods of MIL-STD-810 to demonstrate compliance with 3.1.4:

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a. Humidity - Method 507.2, Procedure I.

b. Salt Fog- Method 509.2, Procedure I.

c. Sand and Dust - Method 510.2, Procedures I and II. For Procedure I, air velocity shall be 1,750 ft/min and high temperature adjustment shall be 125°F. For Procedure II, air velocity shall be 5,700 ft/min; sand concentration of 0.0623 g/ft^3, plus or minus 0.015 g/ft^3; 90 minutes per face duration.

4.6.6.5 Operational checkout. All controls, levers, mechanisms and doors shall be actuated a sufficient number of times to ensure consistency of operation and to determine whether the test exposure produced system degradation.

4.6.7 Reliability demonstration. Compliance with the quantitative reliability requirements of 3.1. 3.1 shall be demonstrated in accordance with task 301 of MIL-STD-781. The test described in 4.6.7.1 shall be continued as necessary to accumulate the required test time.

4.6.7.1 Truck durability test. The vehicle shall be filled with liquid nitrogen from a ground storage tank at the minimum flow rate specified in 3.7.2.1. The truck shall then be driven for ten miles at an average speed of at least 30 mph. During these ten miles, the truck shall complete one full circle left turn and one full circle right turn at maximum cramping angles. The truck shall also make one sudden stop. After completing ten miles, the truck shall be connected to a C-5 nitrogen system mockup. The truck shall supply heated (180° F) gaseous nitrogen to the mockup at the rates specified in 3.7.2.2.1 for a minimum of 30 minutes. The truck shall then supply unheated gaseous nitrogen to the mockup at the rate specified in 3.7.2.2.1 for at least 30 minutes. The remainder of the liquid nitrogen shall then be transfered to the mockup at the rate specified in 3.7.2.2.1. The truck shall be disconnected from the mockup. The truck shall then be reconnected and drain and clear the mockup of all nitrogen. The truck shall transport the nitrogen back to the ground storage tank using the same ten mile route specified above. The nitrogen shall then be transferred to the ground storage tank. This shall complete one cycle.

4.6.8 Maintainability demonstration. Compliance with the maintenance downtime requirements' shall be demonstrated by task performance. All scheduled maintenance tasks and all tasks listed in 3.1.3.2 shall be performed. In addition, corrective maintenance tasks shall be selected by the procuring activity from an approved list of tasks provided by the contractor in the first article test procedure. Not less than 25 percent or more than 50 percent of the total number of tasks shall be demonstrated to verify an average maintenance downtime within the required limit. As part of this demonstration, the recommended frequencies of the scheduled maintenance tasks and the times recorded to accomplish the tasks shall be used to develop an expected value of scheduled maintenance downtime per measure of use, such as calendar time, hours of operation or miles traveled. The capability of performing daily maintenance tasks by operator's wearing arctic mittens shall be demonstrated.

4.6.9 Air transportability analysis. A detailed engineering analysis shall be conducted to demonstrate compliance with 3.1.9. The analysis shall include the tiedowns and all major components and their ability to withstand the acceleration specified in MIL-A-8421. The evaluation shall also include a dimensional analysis for the vehicle while loaded aboard Type C-130 Aircraft.

4.6.10 User test. When required (see 6.2) the vehicle shall be delivered to the Government designated facility for 45 days of operational use. The test period shall begin after the vehicle has been inspected and prepared for use and shall be available for up to 360 hours of operation. The vehicle shall be available for any eight hour period specified by the Government each day, seven days a week. Any period of more than two hours of the same day that the vehicle is out of service for maintenance or repair shall not be considered an available day of the test period. The vehicle shall be operated, and scheduled maintenance shall be performed by the Air Force personnel using the operation and maintenance procedures contained in the preliminary technical data to be supplied with the vehicle. Any parts or

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corrective actions required shall be provided by the contractor; all fuel, agents and nitrogen shall be provided by the Air Force. Test operations will simulate normal Air Force usage and will not exceed any operating parameters defined by this specification. The sole criteria for acceptance/rejection of the user test shall be based on the performance requirements of this specification. The contractor shall maintain a daily log. The log shall be signed by the Government representative and the contractor after each day of activity. The daily log shall document, but not be limited to, the following:

a. Operating time.

b. Tasks completed.

c. Technical manual errors.

d. All failures and corrective actions taken.

e. Operational problems encountered.

f. Government representative comments.

The contractor shall include the user test results, including daily logs of activities, in the first article test report. All systems and subsystems may be tested at the beginning and end of the test period. The contractor shall be responsible for recording all data required for the first article test report.

4.7 Refurbishment. After successful completion of all required testing, each vehicle shall be restored to like new condition.

5. PACKAGING

5.1 Preparation for delivery shall be in accordance with the terms of the contract.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The liquid nitrogen servicing truck covered by this specification is intended for the use of servicing liquid nitrogen systems on the C-5 Aircraft. It shall be capable of purging, filling and saturating the C-5 system with gaseous nitrogen and filling with liquid nitrogen without utilizing an external source of power. It is also capable of self-purging and receiving liquid nitrogen from a ground storage facility.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.

b. Finish color required, if not Forest Green, Color Number 24052 of FED-STD-595 (see 3.8.1.1.1).

c. Identification markings required (see 3.8.2.1).

d. Rustproofing, if other than the manufacturer's standard commercial practice, is required (see 3.8.1.1.3) .

e. User test facility when required (see 4.6.10).

6.3 Data requirements. The following Data Item Descriptions (DID's) must be listed, as applicable, on the Contract Data Requirement List (DD Form 1423) when this specification is applied on a contract, in order to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

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Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4.3.1	DI-T-4902	First Article Inspection Report	See DD Form 1423

The above DID's were those cleared as of the date of this specification. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Definitions.

6.4.1 Belgian Block. A Belgian Block course is paved with unevenly laid granite or basalt block, forming an undulating surface. It duplicates a rough cobblestone road such as is found in many parts of the world. A cobblestone course may be used as an alternate.

6.4.2 Commercial. A manufacturer's standard product, having been marketed, in production and sold in substantial quantities to the general public for a minimum of one year prior to the date of invitation for bid or request for proposal. An item is "sold in substantial quantities" only when the quantities regularly sold are sufficient to constitute a real commercial market. Nominal quantities, such as models, samples, prototypes or experimental units, do not meet this requirement. The "general public" is a significant number of buyers other than the Government or affiliates of the offeror; the item involved must not be for Government end use. For the purpose of this specification, items acquired for "Government end use" include items acquired for foreign military sales.

6.4.3 Cross country terrain. Cross country terrain consists of open fields, broken ground and uneven terrain in general.

6.4.4 Full circle turn at maximum cramping angle. A complete 360° circle within 130% of the vehicle's curb to curb turning diameter(see 3.2.1(i)).

6.4.5 Sudden stop. A complete stop at a deceleration rate of at least 12 ft/sec^2.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the items(s) should be a first article sample, a first production item, or a standard production item from the contractor's current inventory and the number of items to be tested as specified in 4.4. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.6 Subject term (key word) listing.

C-5 aircraft
Fire suppression system
Heat exchanger
Storage dewar

Custodians:
Air Force - 99

Preparing activity:
Air Force - 84

(Project 2320-F332)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-T-83815	2. DOCUMENT DATE (YYMMDD) 90-11-26

3. DOCUMENT TITLE
TRUCK, TANK, LIQUID NITROGEN, AIRCRAFT SERVICING, 900 GALLON

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Michael A. Greene	912-926-5616	468-5616
ADDRESS (Include Zip Code) WR-ALC/LZDTD Robins AFB GA 31098-5365	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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